Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	SWTID-2026-9325
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	Proper Python indentation is maintained throughout the project.
2	Proper File Structure	Files and folders are logically organize	Yes	Project is organized into modules such as main.py, models.py, schemas.py, crud.py, database.py, auth.py, and ai_engine.py.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like monthly_surplus, total_debt, borrower_id, and recommendations clearly indicate their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as calculate_analysis(), generate_negotiation(), get_profile(), and get_repayment_simulation() are descriptive.
5	Code Comments	Inline and block comments explain log	Yes	Comments explain major sections and important logic.
6	Modular Design	Code is split into reusable functions/m	Yes	Code is divided into reusable modules for authentication, CRUD operations, AI engine, database, and schemas.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code is minimized through reusable functions and modular design.
8	Error Handling	Exceptions and errors are handle gracefully	Yes	FastAPI HTTPException is used for validation and error handling.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module (auth.py)	Python, FastAPI	User Registration & Login APIs	High
2	CRUD Module (crud.py)	Python, SQLAlchemy	Loan Management, Profile Management, Negotiation History	High
3	Database Module (database.py)	SQLite, SQLAlchemy	Used across all backend modules	High
4	AI Engine (ai_engine.py)	Python, Google Gemini API	AI Settlement Recommendation & Negotiation Letter Generation	High
5	Settlement Analysis (calculate_analysis())	Python	Financial Dashboard, Debt Analysis & Recommendations	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized modular architecture with separate backend components.
Readability	5	Meaningful naming conventions and consistent coding style improve readability.
Reusability	5	Reusable modules for authentication, CRUD operations, AI services, and database access.
Documentation / Comments	4	Adequate comments explaining important logic; additional documentation could further improve maintainability.
Overall Score	4.8	Well-structured, maintainable, and reusable codebase suitable for an academic AI-based project.